PROGRAM

```
#include <stdio.h>
#include <stdlib.h>
/* Call by Value */
int addByValue(int a, int b)
{
   int res;
   res = a + b;
   return res;
}
/* Call by Reference */
void swapByReference(int *a, int *b)
{
   int temp;
   temp = *a;
   *a = *b;
   *b = temp;
}
int main()
{
   int a, b, res;
   int *ptr;
   int *arr;
   int n, i, sum;
   printf("Enter the first number: ");
   scanf("%d", &a);
   printf("Enter the second number: ");
   scanf("%d", &b);
   /* Pointer demonstration */
```

```
ptr = &a;
printf("\nPointer Demonstration:");
printf("\nValue of a = %d", *ptr);
printf("\nAddress of a = %p", (void *)ptr);
/* Call by Value */
res = addByValue(a, b);
printf("\n\nCall by Value:");
printf("\nSum = %d", res);
/* Call by Reference */
swapByReference(&a, &b);
printf("\n\nCall by Reference:");
printf("\nAfter swapping, first number = %d", a);
printf("\nAfter swapping, second number = %d", b);
/* Dynamic Memory Allocation */
printf("\n\nDynamic Memory Allocation");
printf("\nEnter the number of elements: ");
scanf("%d", &n);
arr = (int *)malloc(n * sizeof(int));
if(arr == NULL) {
   printf("Memory allocation failed.");
   return 0;
}
printf("Enter %d elements:\n", n);
for(i = 0; i < n; i++)
{
   scanf("%d", &arr[i]);
}
sum = 0;
for(i = 0; i < n; i++)
```

```
    {
        sum = sum + arr[i];
    }
    printf("Sum of dynamically allocated array = %d", sum);
    free(arr);
    return 0;
}
```

OUTPUT

```
Enter the first number: 10
Enter the second number: 20
Pointer Demonstration:
Value of a = 10
Address of a = 000000000061FDE4
Call by Value:
Sum = 30
Call by Reference:
After swapping, first number = 20
After swapping, second number = 10
Dynamic Memory Allocation
Enter the number of elements: 5
Enter 5 elements:
1
2
3
4
5
Sum of dynamically allocated array = 15
```